King Saud University
College of Computer & Information Science
CSC111 – Lab Extra
Menus, if-else, switch, and nested loops
All Sections

Lab Exercise 1

We will write a java program to draw shapes using a menu. Here is how the menu will look like:

```
*********************************************************
* 0- Currently drawing '*'. Change it?                  *
* 1- Rectangle.           2- Hollow rectangle.          *
* 3- Right triangle.      4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.            *
* 9- Exit.                                              *
*********************************************************
```

After completing a task, the menu will show up again so the user can select a new task. Name your class **Drawing**.

Note: *use a switch to complete the menu selection.*

Note: *the default printing character is '*'.*

Option 0

When the user selects 0, ask the user to type the new drawing character. Then change the character we use for printing.

Sample Run

```
*********************************************************
* 0- Currently drawing '*'. Change it?                  *
* 1- Rectangle.           2- Hollow rectangle.          *
* 3- Right triangle.      4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.            *
* 9- Exit.                                              *
*********************************************************
==> 0
Enter the new drawing char: A
*********************************************************
* 0- Currently drawing 'A'. Change it?                  *
* 1- Rectangle.           2- Hollow rectangle.          *
* 3- Right triangle.      4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.            *
* 9- Exit.                                              *
*********************************************************
```

Options 1 and 2

If the user selects options 1 or 2, ask the user to enter the *height* and *width* of the rectangle. **In option 1**, use the *height* as the number of lines and *width* as the number of characters that will be printed in each line. **In option 2**, only the first and last lines will be the same as option 1. The rest will have spaces in between the drawing characters.

Sample Run

```
*********************************************************
* 0- Currently drawing '&'. Change it?                  *
* 1- Rectangle.           2- Hollow rectangle.          *
* 3- Right triangle.      4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.            *
* 9- Exit.                                              *
*********************************************************
=> 1
Enter the height: 4
Enter the width: 10
&&&&&&&&&&
&&&&&&&&&&
&&&&&&&&&&
&&&&&&&&&&
*********************************************************
* 0- Currently drawing '&'. Change it?                  *
* 1- Rectangle.           2- Hollow rectangle.          *
* 3- Right triangle.      4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.            *
* 9- Exit.                                              *
*********************************************************
=> 2
Enter the height: 4
Enter the width: 10
&&&&&&&&&&
&        &
&        &
&&&&&&&&&&
```

Options 3 and 4

If the user selects options 3 or 4, ask the user to enter the *height* of the right triangle. **In option 3**, use the *height* as the number of lines where in each line you print the drawing character one more than the previous line. **In option 4**, only the first, second, and last lines will be the same as option 3. The rest will have spaces in between the drawing characters.

Sample Run

```
********************************************************
* 0- Currently drawing '#'. Change it?                 *
* 1- Rectangle.          2- Hollow rectangle.          *
* 3- Right triangle.     4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.            8- Hollow diamond.            *
* 9- Exit.                                             *
********************************************************
=> 3
Enter the height: 5
#
##
###
####
#####
********************************************************
* 0- Currently drawing '#'. Change it?                 *
* 1- Rectangle.          2- Hollow rectangle.          *
* 3- Right triangle.     4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.            8- Hollow diamond.            *
* 9- Exit.                                             *
********************************************************
=> 4
Enter the height: 5
#
##
# #
#  #
#####
```

Options 5 and 6

If the user selects options 5 or 6, ask the user to enter the *height* of the isosceles triangle. **In option 5**, use the *height* as the number of lines where in each line you print the drawing character two more times than the previous line. You will also need to print spaces before the drawing character to fix the triangle. **In option 6**, only the first and last lines will be the same as option 5. The rest will have spaces in between the drawing characters.

Sample Run

```
********************************************************
* 0- Currently drawing 'Q'. Change it?                 *
* 1- Rectangle.           2- Hollow rectangle.         *
* 3- Right triangle.      4- Hollow right triangle.    *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.           *
* 9- Exit.                                             *
********************************************************
=> 5
Enter the height: 5
    Q
   QQQ
  QQQQQ
 QQQQQQQ
QQQQQQQQQ
********************************************************
* 0- Currently drawing 'Q'. Change it?                 *
* 1- Rectangle.           2- Hollow rectangle.         *
* 3- Right triangle.      4- Hollow right triangle.    *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.             8- Hollow diamond.           *
* 9- Exit.                                             *
********************************************************
=> 6
Enter the height: 5
    Q
   Q Q
  Q   Q
 Q     Q
QQQQQQQQQ
```

Options 7 and 8

If the user selects options 7 or 8, ask the user to enter the *height* of the diamond. **In option 7**, it's the same as option 5, you draw an isosceles triangle with half the *height* then draw an upside down isosceles triangle with the other half. **In option 8**, only the first and last lines will be the same as option 7. The rest will have spaces in between the drawing characters.

***Note:** If the height is an odd number, the two triangles will have different heights. For example: if the height is 5, the normal triangle will have a height of 3 while the upside down triangle will have a height of 2. You can use the method **Math.ceil**.*

```
Math.ceil(5 / 2.0) ⇒ Math.ceil(2.5) ⇒ 3
```

Sample Run

```
********************************************************
* 0- Currently drawing 'M'. Change it?                 *
* 1- Rectangle.          2- Hollow rectangle.          *
* 3- Right triangle.     4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.            8- Hollow diamond.            *
* 9- Exit.                                             *
********************************************************
=> 7
Enter the height: 6
  M
 MMM
MMMMM
MMMMM
 MMM
  M
********************************************************
* 0- Currently drawing 'M'. Change it?                 *
* 1- Rectangle.          2- Hollow rectangle.          *
* 3- Right triangle.     4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.            8- Hollow diamond.            *
* 9- Exit.                                             *
********************************************************
```

```
=> 7
Enter the height: 5
  M
 MMM
MMMMM
 MMM
  M
********************************************************
* 0- Currently drawing 'M'. Change it?                 *
* 1- Rectangle.          2- Hollow rectangle.          *
* 3- Right triangle.     4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.            8- Hollow diamond.            *
* 9- Exit.                                             *
********************************************************
=> 8
Enter the height: 6
  M
 M M
M   M
M   M
 M M
  M
********************************************************
* 0- Currently drawing 'M'. Change it?                 *
* 1- Rectangle.          2- Hollow rectangle.          *
* 3- Right triangle.     4- Hollow right triangle.     *
* 5- Isosceles triangle. 6- Hollow isosceles triangle. *
* 7- Diamond.            8- Hollow diamond.            *
* 9- Exit.                                             *
********************************************************
=> 8
Enter the height: 5
  M
 M M
M   M
 M M
  M
```

Done…